One entry, formed in full

4
ENTRIES

Row 1 of A meets column 2 of B: three products, added, give the entry 19. Each of the other three entries is the same operation on a different row and column.

A 2 × 3 · B 3 × 2 → AB 2×2

The inner pair matches, then vanishes. Only the outer numbers survive into the result.

$$\begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & 4 \end{bmatrix} \cdot \begin{bmatrix} 5 & 1 \\ 2 & -3 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} 5 & \mathbf{19} \\ 8 & 29 \end{bmatrix}$$

A 2×3 · B 3×2 · AB 2×2

HOW THAT ENTRY WAS FORMED

c12 = 1 · 1 + 0 · -3 + 3 · 6 = 1 + 0 + 18 = **19**

Row 1 of A pairs off against column 2 of B, term by term, and the 3 products are summed. The shared length 3 is the inner dimension — it is consumed by the sum and does not appear in the answer.

MULTIPLICATIONS	ADDITIONS	REVERSE PRODUCT
12	**8**	**3×3**
m · n · p — one per term of every entry	m(n−1)p — one fewer than the terms	defined here, and a different size

WHAT IT COSTS

Each entry takes $n = 3$ multiplications and 2 additions, and there are $m \times p = 4$ entries. Straight from the definition an $m \times n$ by $n \times p$ product takes mnp multiplications: 12 here.